

# Oluwatobi Melvyn Mayungbo

Data Scientist, Machine Learning and Statistics

mlvyn.t@gmail.com | [LinkedIn: linkedin.com/in/oluwatobi-mayungbo-3a567026b](https://www.linkedin.com/in/oluwatobi-mayungbo-3a567026b) | [GitHub: github.com/MelvTheGoat](https://github.com/MelvTheGoat)

## SUMMARY

---

Data scientist with a B.Sc. in Statistics who writes production code, not just notebooks. Built time-series pipelines in Python and SQL with Prefect, dbt and DuckDB, sequence models over event histories in PyTorch, and rule-based and ML scoring systems. Tests ideas as hypotheses and ships with Docker.

## SKILLS

---

**Engineering and MLOps:** Python, SQL, Prefect, dbt-core, DuckDB, Docker, PostgreSQL, Redis, FastAPI, GCP Cloud Run, AWS ECS Fargate, Railway, MLflow, Evidently, CI/CD (GitHub Actions), Git, Pytest, Streamlit, React

**Machine Learning:** Scikit-learn, PyTorch, Sequence Models (GRU, TCN, Transformers), Feature Engineering, LightGBM, XGBoost, Deep Learning, Model Calibration (Isotonic Regression, Temperature Scaling), SHAP

**Statistics and Causal Inference:** Time-Series Backtesting, Statistical Inference, Experiment Design, Power Analysis, CUPED, Causal Inference (S/T/X-Learners, Causal Forests), Uplift Modeling, Qini and AUUC, Logistic Regression, WOE Scorecards

**Data Analysis:** Pandas, NumPy, SQL, Anomaly Detection

**Payments and Risk:** Transaction Fraud Detection, Credit Risk Decisioning, Payment Reconciliation, Double-Entry Ledgers and Idempotency, Regulatory Compliance (CBN/NDPA), Demand Forecasting, Web3 Fraud Risk Scoring

**AI and LLM Systems:** Retrieval-Augmented Generation (RAG), Hybrid Search (BM25, Dense Embeddings, RRF), LLM Evaluation and Guardrails, Prompt Engineering, Citation Verification, Model Context Protocol (MCP), Hugging Face, ONNX Runtime

## PROJECTS

---

### Self-Operating Demand Forecasting Platform | [GitHub](#)

*Tech: Python, Prefect, dbt-core, DuckDB, LightGBM, MLflow, Evidently, FastAPI, Docker, GitHub Actions*

- Built an end-to-end hourly demand forecasting system for NYC taxi zones, producing 24-hour quantile forecasts through an orchestrated Prefect pipeline spanning ingestion, dbt/DuckDB transformations, backtesting, drift monitoring and conditional retraining.
- Validated with rolling-origin backtesting across 6 expanding-window folds against seasonal-naive and historical-mean baselines, reporting MASE and pinball loss, with automated leakage tests that fail the build if any feature reads past the forecast origin.
- Modelled asymmetric business cost by penalising unmet demand at 3× idle supply and shipping the cost-optimal quantile rather than the median, cutting expected cost by 25% on an identical model and beating a seasonal-naive baseline by 32%.

### Sequence-Based Transaction Fraud Detection System | [GitHub](#)

*Tech: Python, PyTorch, LightGBM, ONNX Runtime, FastAPI, Docker, SHAP*

- Built and compared three deep sequence architectures (GRU, Temporal Convolutional Network and a causally masked Transformer encoder) over customer transaction histories, with learned embeddings for high-cardinality merchant (~2.9k vocabulary), MCC and device-type categoricals; all models under 115k parameters and CPU-trainable in under 6 minutes.
- Benchmarked all architectures against a strong LightGBM baseline on engineered velocity and deviation features, reporting PR-AUC and precision at fixed review capacity (top 0.1%, 0.5%, 1%) across 3 seeds, with a per-attack-type breakdown covering card testing, account takeover and merchant compromise; sequence models cut expected cost per transaction 27% under an explicit cost model.

### web3-risk-mcp: Web3 Fraud Risk MCP Server | [GitHub](#)

*Tech: Python, Model Context Protocol (MCP), Etherscan V2, GoPlus, DexScreener, Docker, GitHub Actions*

- Designed a rule-based, fully explainable scorer where findings carry stable IDs and decisive findings set a floor of 75 so trust signals can never hide them; each score carries a confidence level based on how many sources answered.

### Premier League Match Predictor | [GitHub](#)

*Tech: Python, LightGBM, scikit-learn, pandas, SQLite, Flask, Vercel*

- Built a predictor that forecasts Premier League outcomes and scorelines each gameweek and retrains itself after every gameweek, keeping a public record of every past prediction.
- Backtested gameweek by gameweek over 1,050 matches: 52.1% outcome accuracy against 43.2% for always picking the home side, and log loss 0.9948 against 1.0061 for an Elo-only baseline, with well calibrated probabilities.

## EXPERIENCE

---

### Machine Learning Instructor, SQI College of ICT

March 2026 to Present

- Teach machine learning and deep learning to student cohorts, covering statistical foundations, classical ML algorithms, neural architectures, sequence models and production evaluation metrics.
- Instruct students on data analysis, feature engineering and model building with PyTorch, Scikit-Learn, Pandas and NumPy.

### Internal Auditor, NISER

May 2024 to July 2024

- Analysed financial and operational records to identify inconsistencies and irregularities, performing manual anomaly detection across complex transactional datasets.

## EDUCATION

---

### B.Sc. Statistics, University of Ibadan

January 2021 to March 2025

### Professional Diploma in Artificial Intelligence, SQI College of ICT

July 2025 to Present

## CERTIFICATIONS

---

- Google Data Analytics Professional Certificate
- Google Advanced Data Analytics Professional Certificate
- Claude Code In Action